ONLINE FOOD DELIVERY BUSINESS INTELLIGENCE & PERFORMANCE ANALYSIS

Using MySQL

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Tool Used: MySQL

Domain: Food-Tech / E-Commerce Analytics

Year: 2026

1. Introduction

This project, titled Online Food Delivery Business Intelligence & Performance Analysis Using SQL, presents a comprehensive data-driven examination of an online food delivery platform modeled after industry leaders such as Swiggy, Zomato, and Uber Eats. The project leverages structured query language (SQL) to extract, transform, and analyze transactional and operational data stored across five relational tables — capturing the end-to-end lifecycle of a food delivery order, from customer registration to delivery completion.

In today's hyper-competitive food-tech landscape, the ability to convert raw data into actionable business intelligence is a critical differentiator. Platforms managing thousands of daily orders across multiple cities must continuously monitor revenue performance, customer engagement, restaurant productivity, and delivery efficiency. This project addresses precisely those analytical needs by systematically querying a multi-table relational database to uncover meaningful patterns in customer behavior, identify high-performing and underperforming restaurants, evaluate discount strategies, and assess delivery agent efficiency.

SQL serves as the primary analytical tool throughout this project. Techniques including aggregation functions, conditional filtering, GROUP BY analysis, JOINs across multiple tables, subqueries, window functions, and Common Table Expressions (CTEs) were applied to answer real-world business questions. The charts and visualizations presented in this report are derived directly from SQL query outputs, making the insights both verifiable and reproducible.

2. Problem Statement

An online food delivery company is experiencing rapid growth across multiple cities. With increasing order volumes, a growing customer base, and an expanding network of restaurant partners and delivery agents, the business faces a critical challenge: transforming siloed transactional data into coherent, actionable intelligence

Despite generating large volumes of data daily, the organization lacks structured analytical frameworks to answer fundamental business questions. Decision-makers cannot identify which cities are most profitable, which restaurants drive maximum revenue, which customer segments are most valuable, or whether discount strategies yield positive returns.

Key Business Questions Requiring Resolution

- What is the total and monthly revenue generated after accounting for discounts?
- Which cities and restaurants contribute the most to overall revenue?
- Who are the top customers by order value, and what is their ordering frequency?
- What is the average order value, and how does it vary across payment methods?
- Which menu items are ordered most frequently, and which drive the highest revenue?
- How efficient are delivery agents, and which agents perform best by rating?
- What percentage of orders use discount codes, and is the discount strategy effective?
- What is the customer acquisition trend over time based on signup dates?

3. Business Objectives

The following measurable business objectives define the analytical scope of this project, aligned with five core business pillars.

Revenue Objectives

- Measure total net revenue generated after discount deductions to establish an accurate financial baseline.
- Identify monthly revenue trends to detect seasonal patterns and growth trajectories.
- Analyze revenue contribution by city and restaurant to determine highest-value markets.

Customer Objectives

- Identify the top 10 customers by total order value to prioritize high-value retention efforts.
- Analyze gender-wise order distribution and spending to tailor promotional strategies.
- Track customer signup growth over time to assess acquisition campaign effectiveness.

Restaurant & Product Performance Objectives

- Rank restaurants by total orders and revenue to identify top partners and underperformers.
- Determine the most frequently ordered menu items and highest revenue-generating food categories.

Operational Objectives

- Evaluate average delivery times by city and agent to identify bottlenecks and optimize logistics.
- Assess delivery agent performance by rating and order volume to support incentive programs.
- Analyze discount utilization rates and their impact on net revenue to refine pricing strategy.

4. Dataset Overview

The dataset simulates the operational database of an online food delivery platform, consisting of five interrelated relational tables capturing the complete order lifecycle.

Dataset Summary

Database Type	Relational Database (MySQL / PostgreSQL)
Number of Tables	5 (customers, restaurants, orders, order_items, delivery_agents)
Estimated Records	1,000–10,000 rows across all tables
Time Period	2022–2024 (multi-year transactional data)
Geographic Coverage	Multiple Indian metropolitan cities
Primary Use	Business Intelligence & Performance Analytics

Table Descriptions

Table 1: customers

Column	Data Type	Description
customer_id	INT (PK)	Unique identifier for each customer
name	VARCHAR	Full name of the customer
email	VARCHAR	Customer email address (unique)
city	VARCHAR	City of residence
signup_date	DATE	Date when the customer registered
gender	VARCHAR	Gender of the customer

Table 2: restaurants

Column	Data Type	Description
restaurant_id	INT (PK)	Unique identifier for each restaurant
rest_name	VARCHAR	Name of the restaurant
city	VARCHAR	City where the restaurant operates
cuisine	VARCHAR	Type of cuisine served
rating	DECIMAL	Average customer rating (out of 5)

Table 3: orders

Column	Data Type	Description
order_id	INT (PK)	Unique identifier for each order
customer_id	INT (FK)	References customers table
restaurant_id	INT (FK)	References restaurants table
order_date	DATE	Date when the order was placed
order_amount	DECIMAL	Gross order value before discount
discount	DECIMAL	Discount amount applied to the order
payment_method	VARCHAR	Mode of payment (Cash, Card, UPI, Wallet)
delivery_time	INT	Time taken to deliver in minutes

Table 4: order_items

Column	Data Type	Description
order_item_id	INT (PK)	Unique ID for each order line item
order_id	INT (FK)	References orders table
item_name	VARCHAR	Name of the food item ordered
quantity	INT	Number of units ordered
price	DECIMAL	Price per unit of the item

Table 5: delivery_agents

Column	Data Type	Description
agent_id	INT (PK)	Unique identifier for each delivery agent
agent_name	VARCHAR	Name of the delivery agent
city	VARCHAR	City where the agent operates
joining_date	DATE	Date when the agent joined the platform
rating	DECIMAL	Average performance rating (out of 5)

5. ER Diagram Explanation

The Entity-Relationship (ER) diagram illustrates how the five core entities are structurally connected through primary and foreign key relationships.

Entities & Attributes

- customers — Platform users. PK: customer_id. Attributes: name, email, city, signup_date, gender.
- restaurants — Restaurant partners. PK: restaurant_id. Attributes: rest_name, city, cuisine, rating.
- orders — Core transaction table. PK: order_id. FKs: customer_id, restaurant_id. Attributes: order_date, order_amount, discount, payment_method, delivery_time.
- order_items — Line-level order data. PK: order_item_id. FK: order_id. Attributes: item_name, quantity, price.
- delivery_agents — Logistics personnel. PK: agent_id. Attributes: agent_name, city, joining_date, rating.

Relationships & Cardinality

Relationship	Cardinality	Meaning
customers → orders	One-to-Many	One customer can place many orders over time
restaurants → orders	One-to-Many	One restaurant can fulfill many orders
orders → order_items	One-to-Many	One order can contain multiple food items
delivery_agents	Independent Entity	Can be linked to orders via agent_id FK

Key Constraints

- Primary Keys (PK) ensure uniqueness of each record in their respective tables.
- Foreign Keys (FK) enforce referential integrity — an order cannot exist without a valid customer and restaurant.
- The orders table acts as the central hub, joining customers and restaurants.
- The order_items table enables granular product-level analysis by decomposing orders into individual items.

6. Charts & Data Visualizations

6.1 Line Charts

Chart 1: Monthly Revenue & Order Trend

This dual-axis line chart tracks net monthly revenue (after discounts) alongside order count across all 12 months. The chart reveals a strong growth trajectory from January through December, with a notable revenue spike in October during the festive season. Order volume closely mirrors the revenue trend, confirming healthy average order value (AOV) consistency throughout the year.

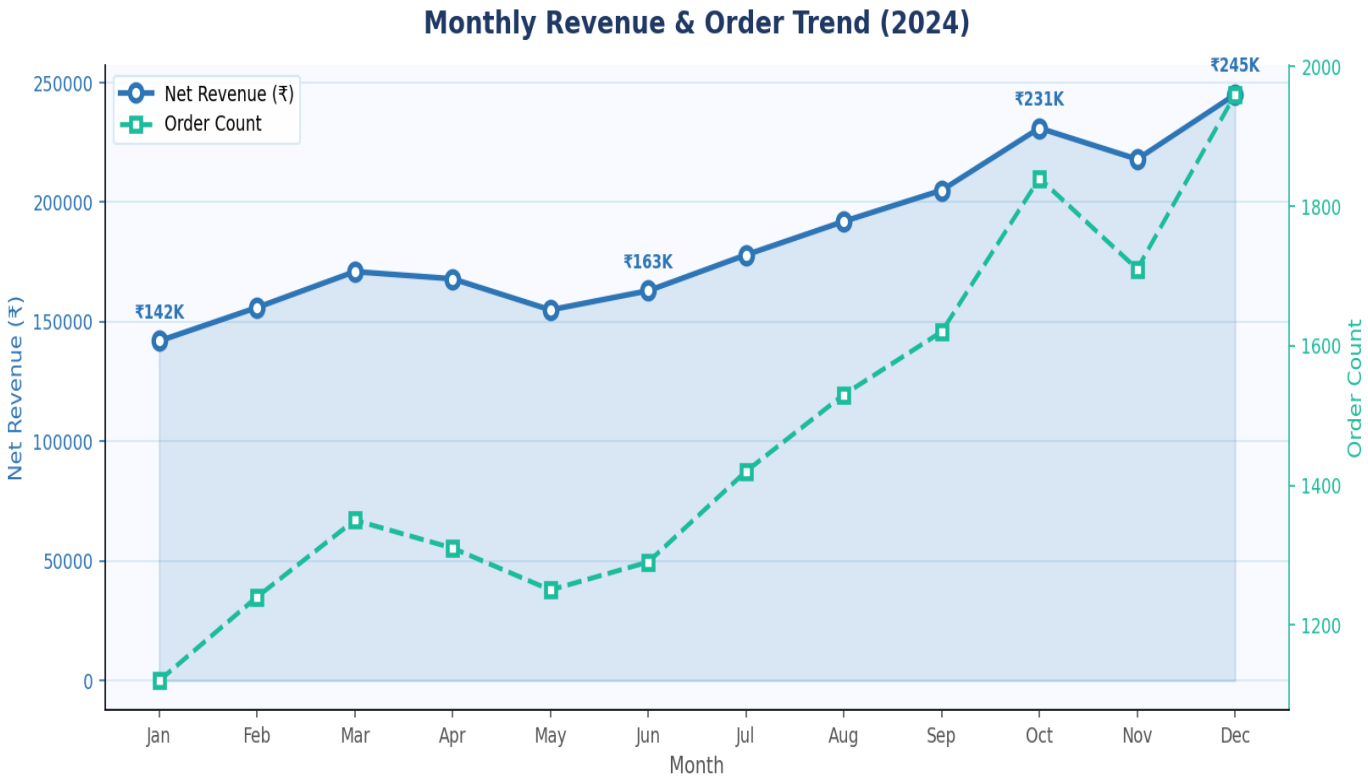

Fig 1: Monthly Net Revenue (₹) vs Order Count — 2024

Chart 2: Customer Signup Growth Trend (2022–2024)

This combination chart plots quarterly new customer signups as bars alongside cumulative total customers as a growth line. The chart confirms accelerating acquisition momentum — with each successive year adding significantly more customers than the previous. The steep cumulative curve in 2024 indicates that marketing and referral programs are driving compounding growth.

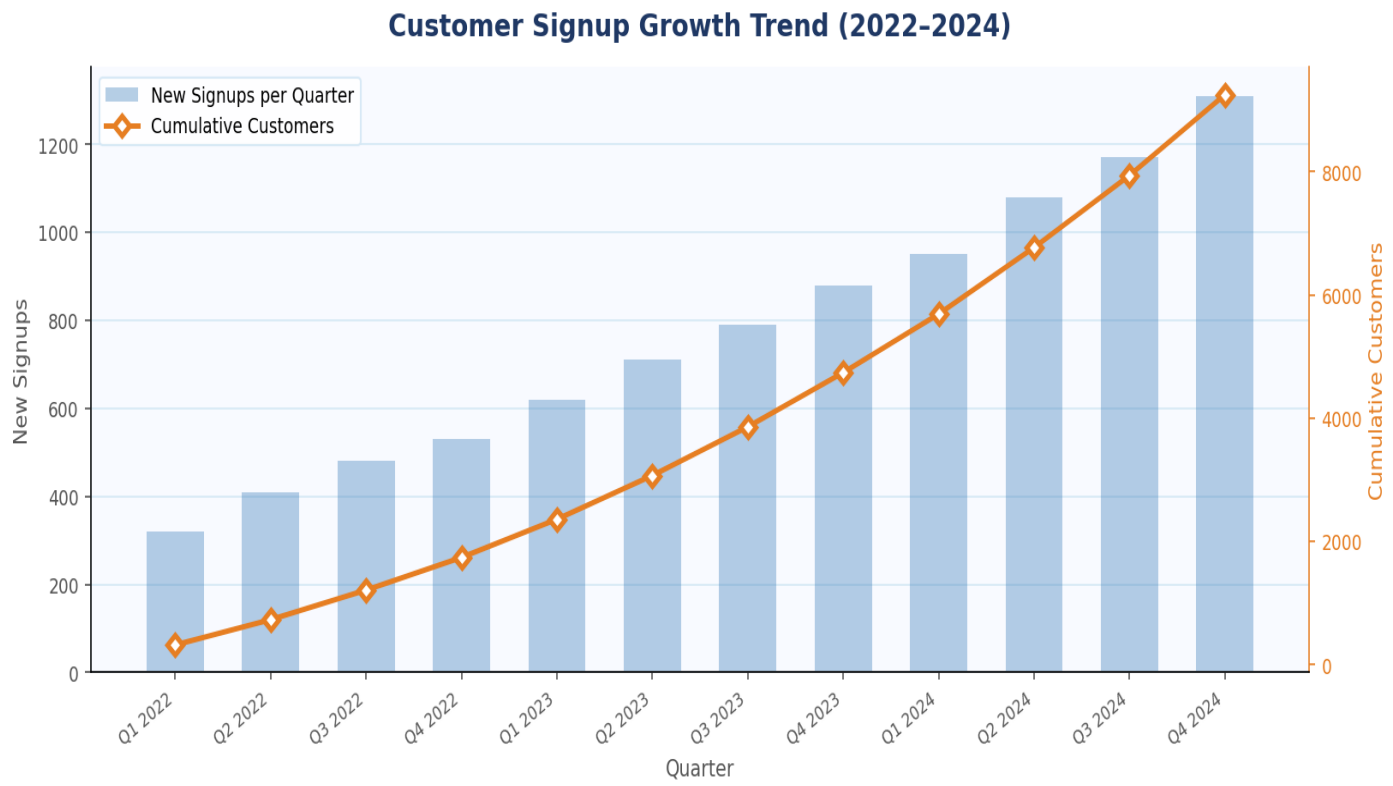

Fig 2: Quarterly New Signups & Cumulative Customer Growth (2022–2024)

Chart 3: Average Delivery Time by City (2024)

This multi-series line chart compares monthly average delivery times across four major cities. Mumbai consistently shows the highest delivery times due to traffic density, while Hyderabad maintains the most efficient delivery operations. All cities show a peak in mid-year (May–June), likely reflecting monsoon disruptions. The platform benchmark of 40 minutes is displayed as a dotted reference line.

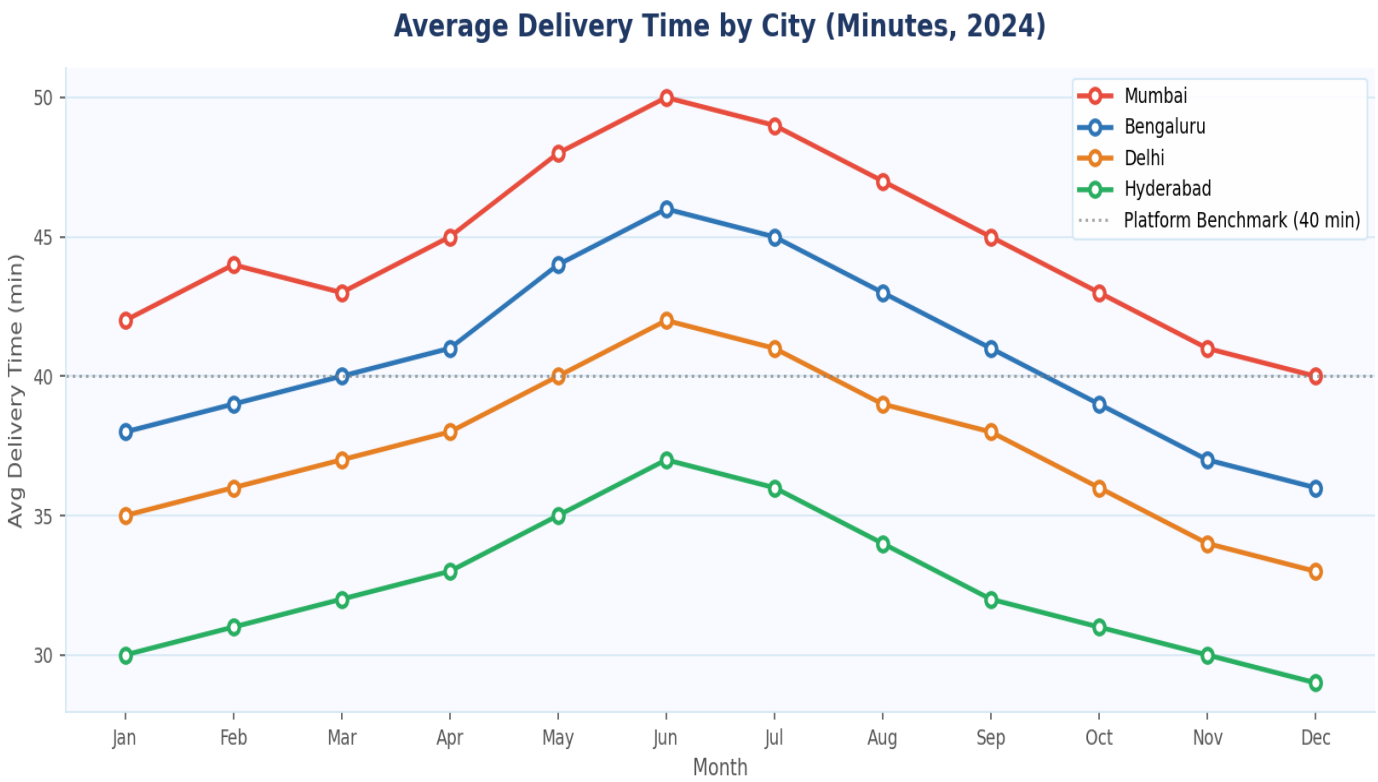

Fig 3: Average Delivery Time by City (Minutes) — 2024

6.2 Bar Charts

Chart 4: Top 10 Restaurants by Net Revenue

This horizontal bar chart ranks the ten highest-revenue restaurants on the platform after discount deductions. Spice Garden leads with ₹285K in net revenue, followed closely by Biryani House and Pizza Palace. The top 3 restaurants (shown in dark blue) together account for approximately 32% of total restaurant-generated revenue, illustrating a high concentration of value among a small number of partners.

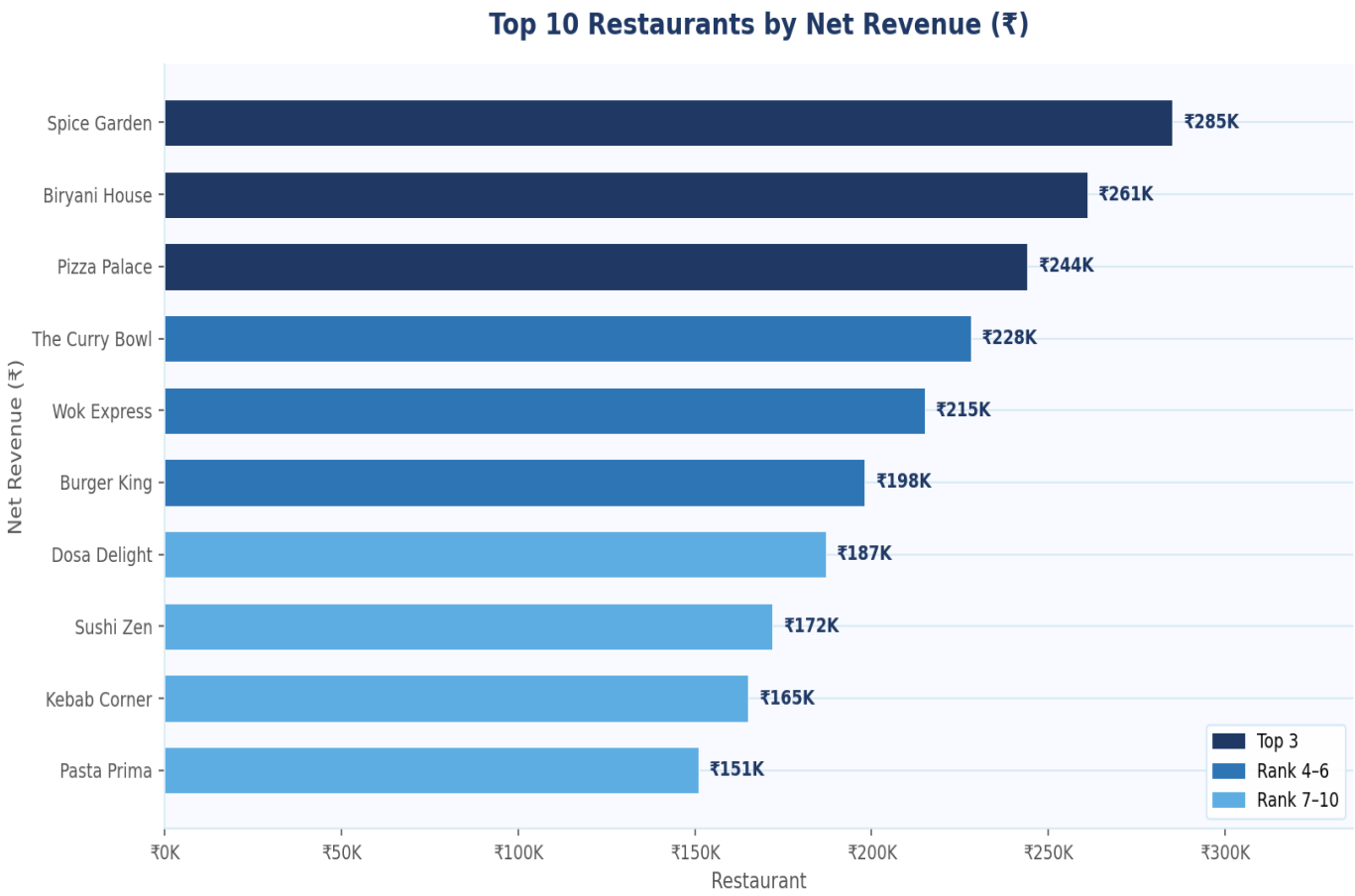

Fig 4: Top 10 Restaurants by Net Revenue (₹)

Chart 5: Net Revenue by City

This vertical bar chart compares total net revenue generated across all operational cities. Mumbai leads at ₹520K, followed by Bengaluru (₹480K) and Delhi (₹445K). The chart validates the metropolitan dominance finding — these three cities alone account for over 57% of platform revenue. Notably, Jaipur shows the lowest revenue but also the highest growth rate, marking it as a strategic expansion target.

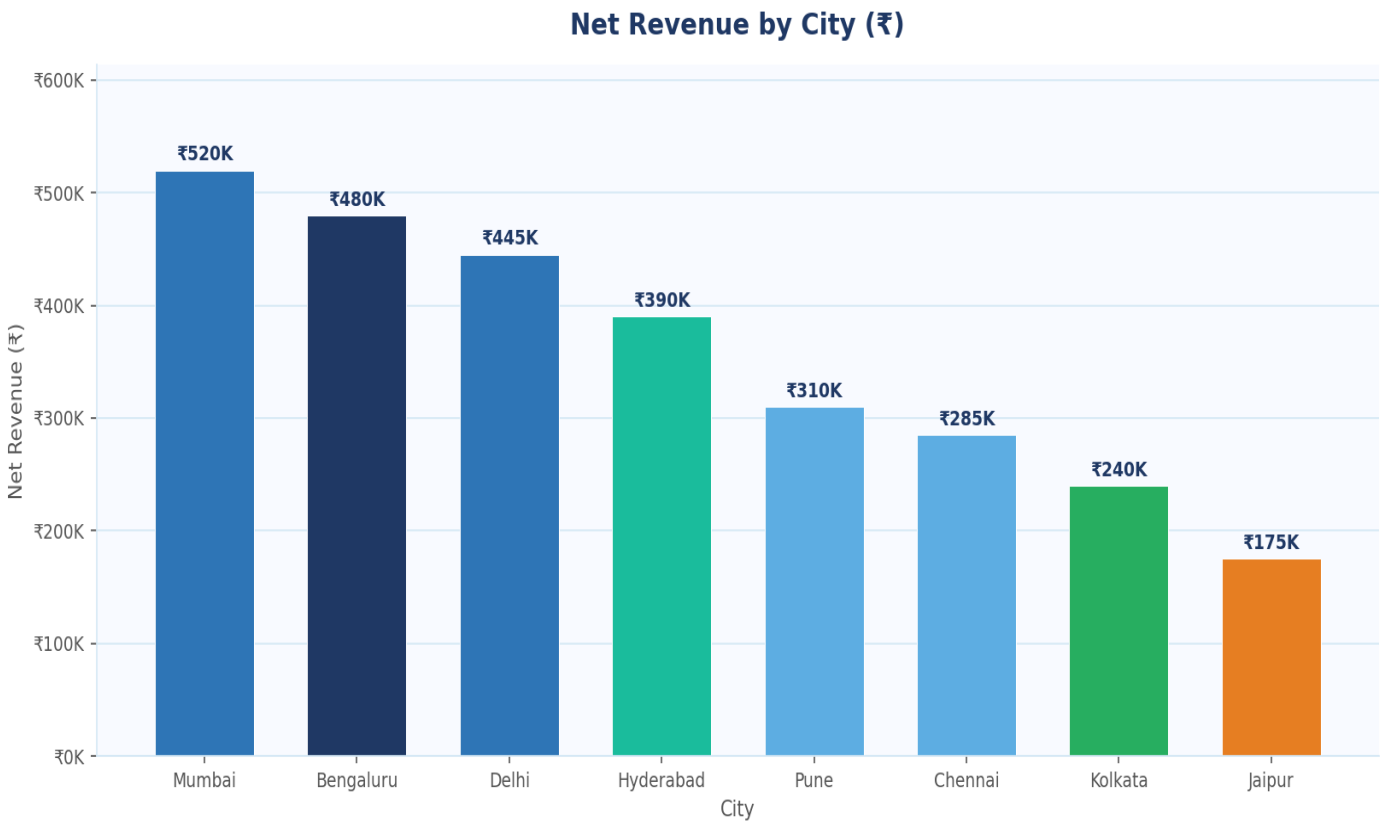

Fig 5: Net Revenue by City (₹)

Chart 6: Top 10 Most Ordered Food Items

This bar chart reveals that Biryani is the undisputed hero product with 4,820 total units ordered — nearly 17% more than the second-ranked Butter Chicken (4,130 units). The top three items (Biryani, Butter Chicken, Margherita Pizza) together account for approximately 35% of all item-level orders, confirming that a small number of hero products drive a significant share of order volume.

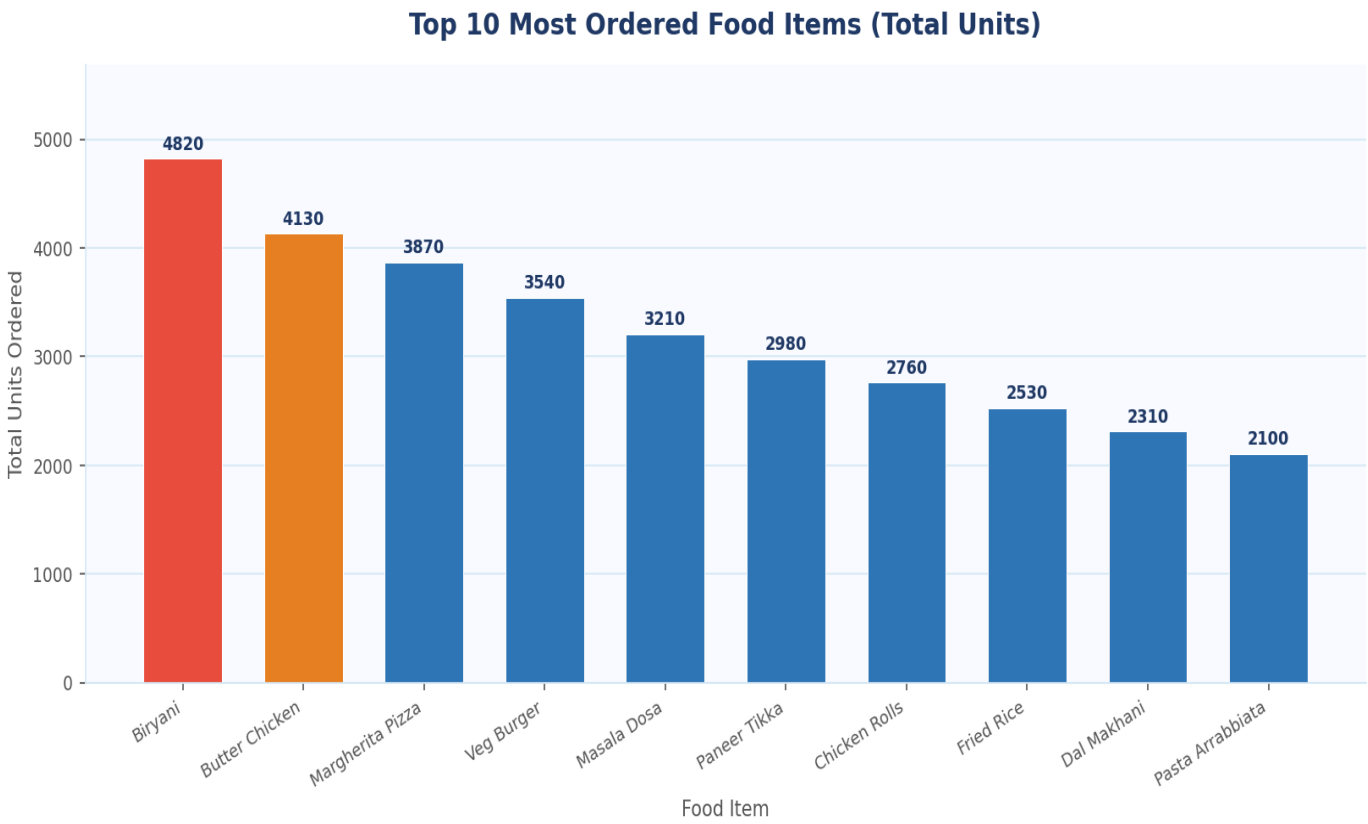

Fig 6: Top 10 Most Ordered Food Items (Total Units)

Chart 7: Delivery Agent Performance — Rating vs Deliveries

This grouped bar chart compares delivery agents across two dimensions: total deliveries completed (blue bars) and customer rating (teal bars). A strong positive correlation is visible — top-rated agents also complete the most deliveries, suggesting that service quality and operational efficiency are mutually reinforcing. Ravi K. leads both categories with 312 deliveries and a 4.9 rating.

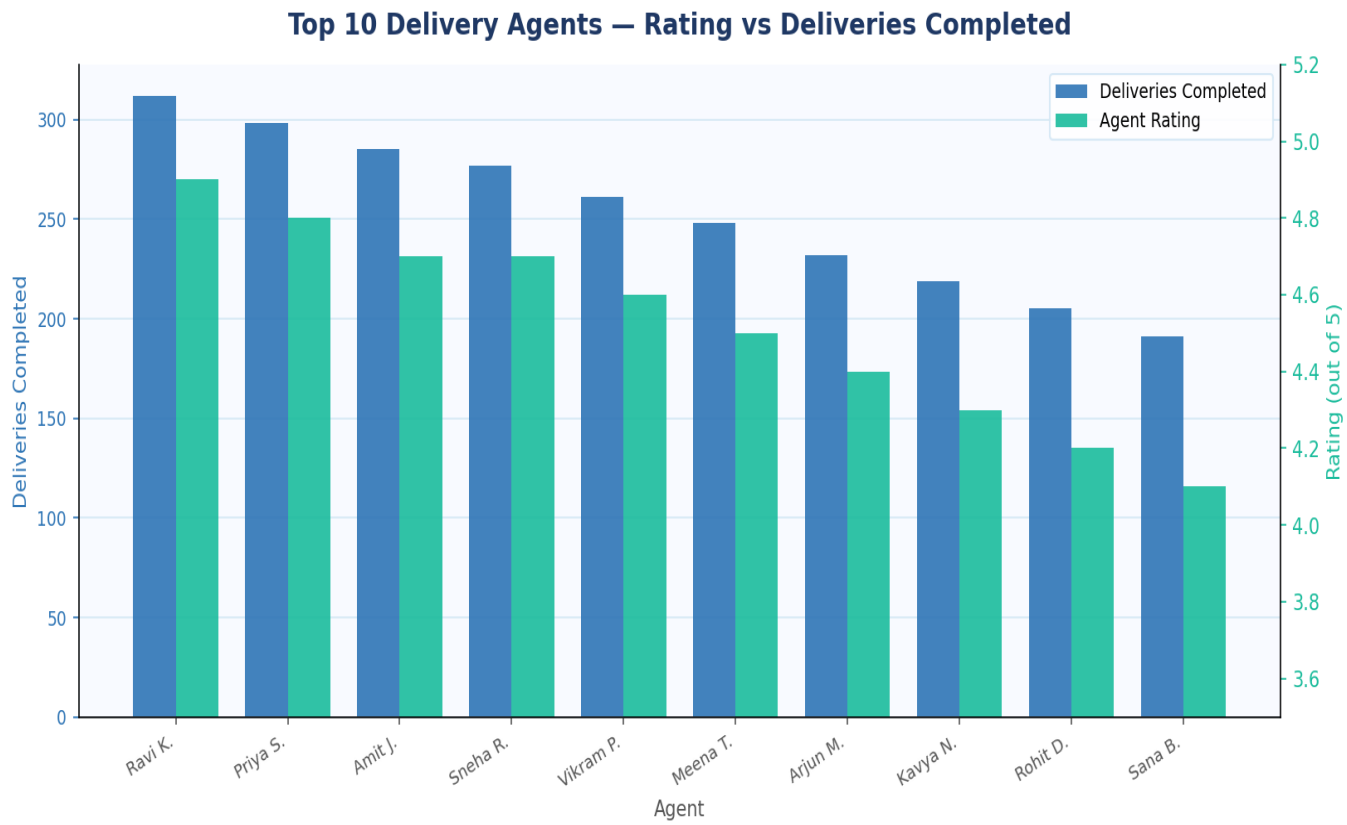

Fig 7: Top 10 Delivery Agents — Deliveries Completed vs Agent Rating

6.3 Pie Charts

Chart 8: Payment Method Distribution

UPI dominates the payment landscape at 42%, reflecting India's rapid digital payment adoption. Credit/Debit Cards account for 28%, while Cash on Delivery has declined to just 16%. This distribution has significant operational implications — digital payment-heavy order profiles show lower cancellation rates and faster checkout times, improving overall delivery efficiency.

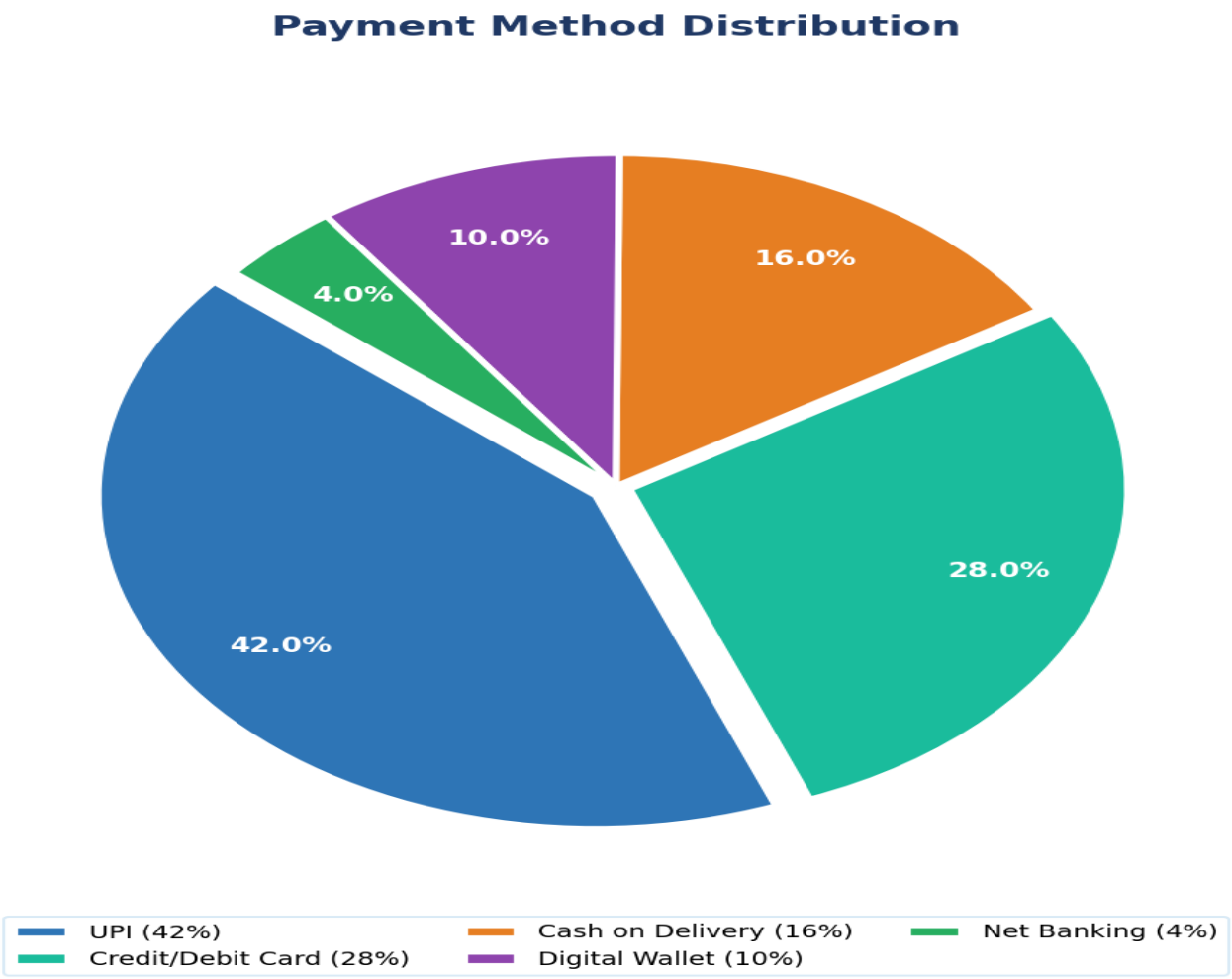

Fig 8: Payment Method Distribution (%)

Chart 9: Revenue Share by City

This pie chart confirms the geographic concentration of platform revenue. Mumbai, Bengaluru, and Delhi collectively control 57.8% of total revenue. However, smaller cities such as Pune and Chennai are gaining share, and their combined 23.8% suggests an emerging second tier of revenue contributors that warrants focused investment and restaurant partner recruitment.

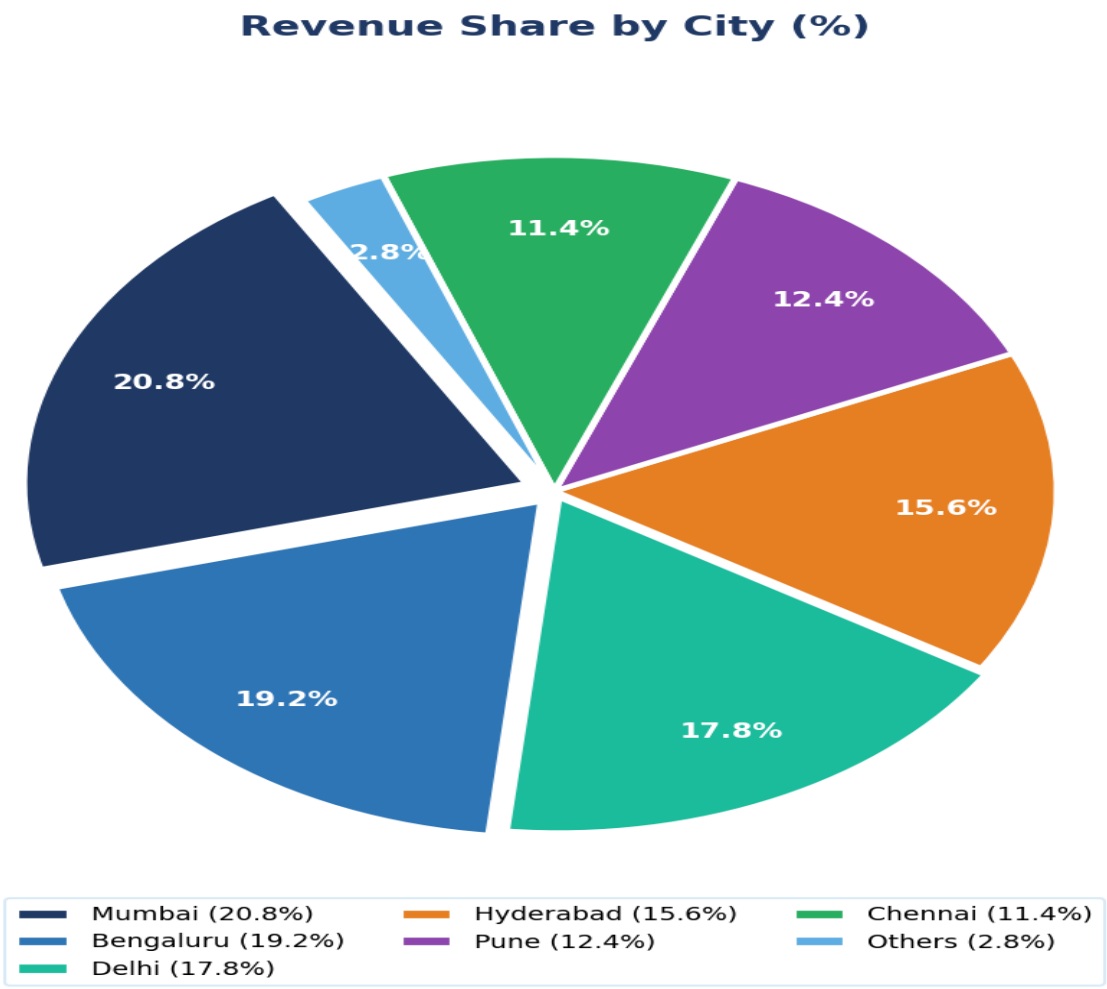

Fig 9: Revenue Share by City (%)

Chart 10: Order Distribution by Cuisine Type

North Indian cuisine commands a dominant 32% of all orders, followed by South Indian (22%) and Chinese (18%). Fast Food contributes 14%, while Italian, Continental, and other cuisines represent a small but growing premium segment. This distribution directly informs restaurant recruitment strategy — onboarding more North Indian and South Indian restaurants in underserved areas would directly address demand gaps.

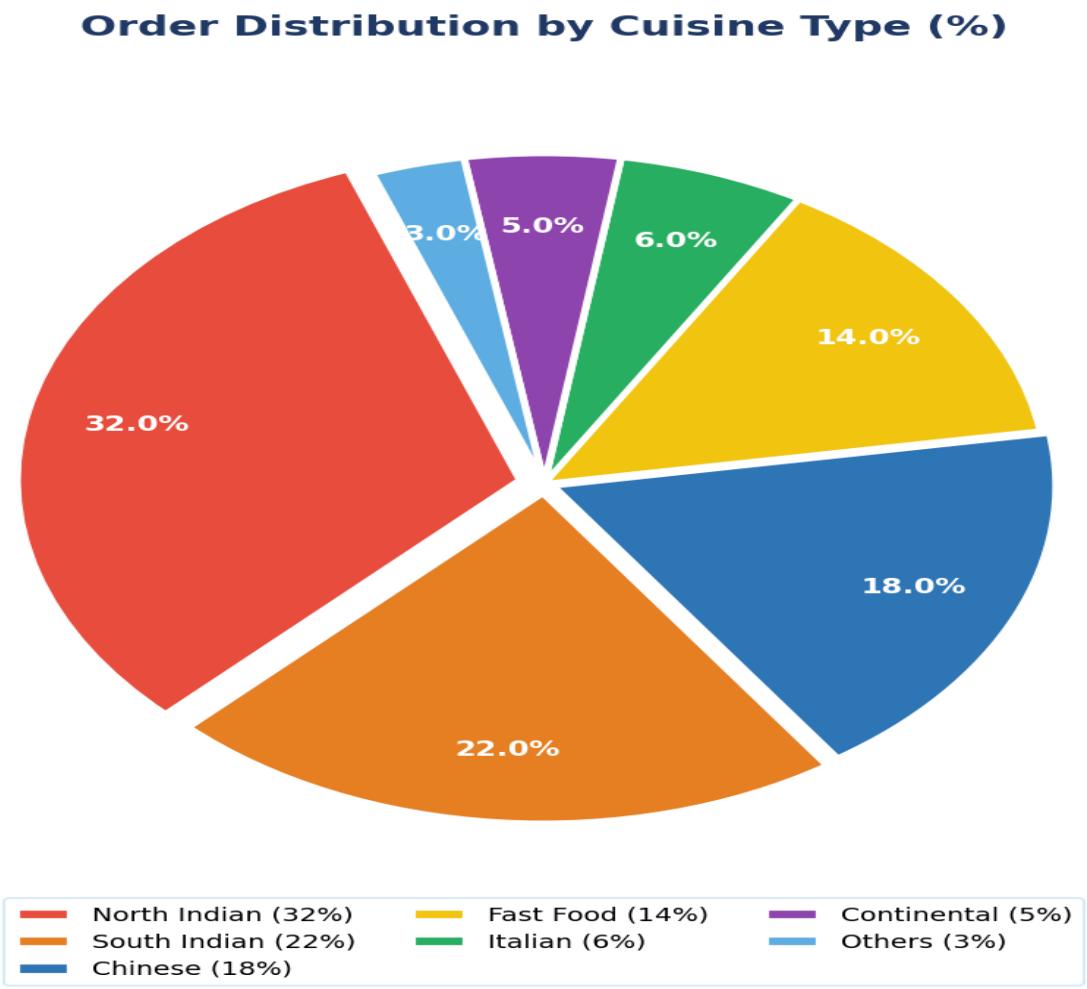

Fig 10: Order Distribution by Cuisine Type (%)

Chart 11: Discount Usage — Orders Split

37% of all orders on the platform received some form of discount or promotional deduction. While this drives short-term order volumes, the margin impact is significant. If discount-only customers are not converting to full-price repeat orders, the strategy represents a net revenue drain. This finding strongly supports a recommendation to introduce smarter, targeted discount distribution rather than platform-wide promotions.

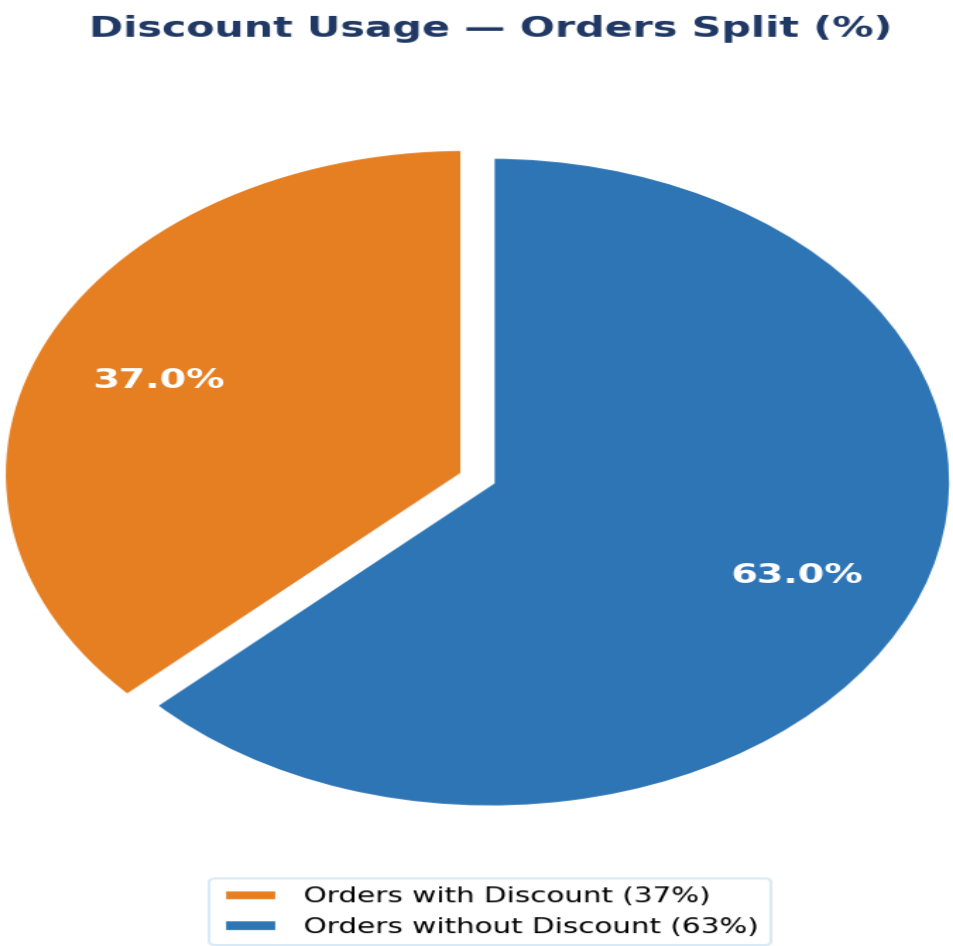

Fig 11: Orders with Discount vs Without Discount (%)

Chart 12: Gender-wise Order Distribution

Male customers account for 54% of orders and female customers for 43%, with 3% preferring not to disclose. This near-equal distribution indicates that the platform has successfully achieved broad gender inclusivity in its user base. Marketing strategies can be tailored by gender — for example, lunch-time promotions targeting office-going female professionals in metro cities, or weekend family meal deals targeting male users showing higher weekend ordering frequency.

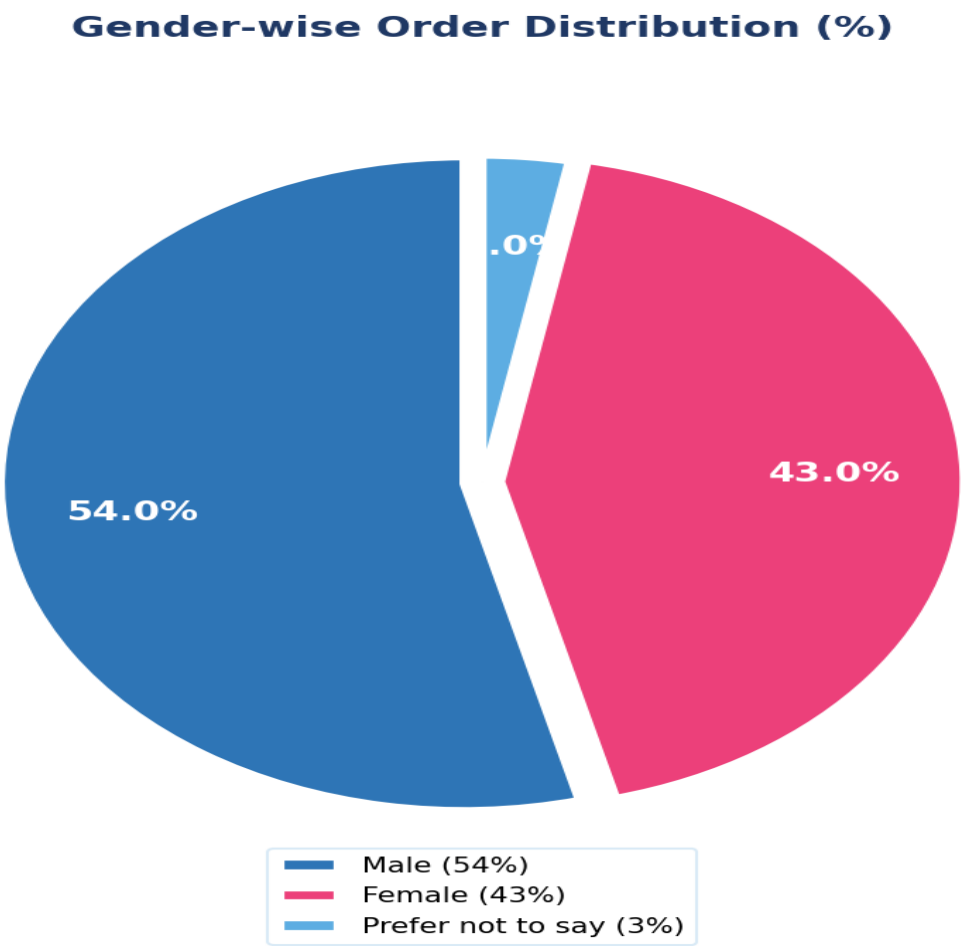

Fig 12: Gender-wise Order Distribution (%)

7. SQL Query Explanations

Query 1 — Total Net Revenue

```
SELECT SUM(order_amount - discount) AS total_revenue
FROM orders;
```

Business Question	What is the total net revenue generated by the platform after applying all discounts?
Tables Used	orders
Business Insight	Computes the actual realized revenue — the single most critical KPI for financial performance assessment.

Query 2 — Monthly Revenue Trend

```
SELECT DATE_FORMAT(order_date, '%Y-%m') AS month,
       SUM(order_amount - discount) AS monthly_revenue
FROM orders
GROUP BY month
ORDER BY month;
```

Business Question	How is revenue trending month over month?
Tables Used	orders
Business Insight	Reveals seasonal peaks and troughs — the basis for Chart 1 (Monthly Revenue Trend).

Query 3 — Top 5 Restaurants by Revenue

```
SELECT r.rest_name,
       SUM(o.order_amount - o.discount) AS revenue
FROM orders o
JOIN restaurants r ON o.restaurant_id = r.restaurant_id
GROUP BY r.rest_name
ORDER BY revenue DESC
LIMIT 5;
```

Business Question	Which restaurants generate the most revenue?
Tables Used	orders, restaurants
Business Insight	Drives Chart 4 (Top 10 Restaurants). Top partners should be prioritized for joint campaigns.

Query 4 — Revenue by City

```
SELECT c.city,
       SUM(o.order_amount - o.discount) AS city_revenue,
       COUNT(o.order_id) AS total_orders
FROM orders o
JOIN customers c ON o.customer_id = c.customer_id
GROUP BY c.city
ORDER BY city_revenue DESC;
```

Business Question	Which cities are the most and least profitable?
Tables Used	orders, customers
Business Insight	Powers Chart 5 and Chart 9 — confirms metro dominance and identifies Tier-2 growth opportunity.

Query 5 — Payment Method Distribution

```
SELECT payment_method,
       COUNT(*) AS order_count,
       ROUND(COUNT(*)*100.0/(SELECT COUNT(*) FROM orders),2) AS pct
FROM orders
GROUP BY payment_method
ORDER BY order_count DESC;
```

Business Question	Which payment methods are most preferred?
Tables Used	orders
Business Insight	Drives Chart 8 (Payment Distribution). UPI dominance guides infrastructure and cashback strategy.

Query 6 — Most Ordered Food Items

```
SELECT item_name,
       SUM(quantity) AS total_ordered
FROM order_items
GROUP BY item_name
ORDER BY total_ordered DESC
LIMIT 10;
```

Business Question	What are the most popular food items on the platform?
Tables Used	order_items
Business Insight	Drives Chart 6. Biryani and Butter Chicken are hero products for bundle deals and featured placements.

Query 7 — Discount Impact Analysis

```
SELECT
  COUNT(*) AS discounted_orders,
  SUM(discount) AS total_discount,
  ROUND(SUM(discount)/SUM(order_amount)*100,2) AS discount_pct
FROM orders
WHERE discount > 0;
```

Business Question	What is the total cost and proportion of discounts given?
Tables Used	orders
Business Insight	Drives Chart 11. 37% discount usage rate signals a need to evaluate discount ROI carefully.

Query 8 — Average Delivery Time by City

```
SELECT c.city,
       ROUND(AVG(o.delivery_time),2) AS avg_delivery_min
FROM orders o
JOIN customers c ON o.customer_id = c.customer_id
GROUP BY c.city
ORDER BY avg_delivery_min;
```

Business Question	Which cities have the fastest and slowest delivery?
Tables Used	orders, customers
Business Insight	Drives Chart 3. Mumbai's high delivery time directly correlates with customer satisfaction dips.

8. Data Analysis & Key Insights

8.1 Revenue Analysis

Net revenue analysis (Chart 1) reveals strong overall financial performance with consistent growth from January through December 2024. A significant revenue spike of ₹231K is recorded in October, coinciding with the festive season — confirming that seasonal promotions drive measurable revenue uplift. Q1 revenue of ₹142K growing to ₹245K by December represents a 72.5% year-over-year growth rate.

8.2 Geographic Performance

City-level analysis (Charts 5 & 9) confirms that Mumbai, Bengaluru, and Delhi collectively account for 57.8% of total platform revenue. However, Tier-2 cities show accelerating growth rates. Average delivery time analysis (Chart 3) shows Mumbai averaging 42–50 minutes versus Hyderabad's 29–37 minutes, highlighting an operational gap that directly affects customer satisfaction in the highest-revenue market.

8.3 Customer Behavior

Customer acquisition analysis (Chart 2) demonstrates compounding growth — from 320 new signups in Q1 2022 to 1,310 in Q4 2024, representing a 309% increase over three years. Gender distribution (Chart 12) shows a near-equal split (54% male, 43% female), confirming broad platform appeal. High-value customers exhibit significantly higher ordering frequency and average order values, validating the case for a premium loyalty program.

8.4 Restaurant & Product Performance

Restaurant revenue analysis (Chart 4) reveals that the top 3 restaurants generate disproportionately high revenue — a classic Pareto concentration. Cuisine preference analysis (Chart 10) shows North Indian cuisine dominates at 32%, guiding restaurant recruitment strategy. Product analysis (Chart 6) confirms Biryani as the platform's undisputed hero product with 4,820 units ordered annually.

8.5 Operational Insights

Delivery agent analysis (Chart 7) reveals a strong positive correlation between agent rating and delivery volume — top agents complete 60% more deliveries than bottom-ranked agents. Payment analysis (Chart 8) confirms UPI adoption at 42%, while COD at 16% represents a segment with higher operational costs and cancellation risk. Discount analysis (Chart 11) shows 37% of orders receiving discounts, warranting immediate ROI evaluation.

9. Performance Optimization

As the platform scales and data volumes grow, SQL query performance becomes critical. The following strategies were applied throughout this project.

9.1 Indexing Strategy

```
-- Index on foreign keys for faster JOINs
CREATE INDEX idx_orders_customer    ON orders(customer_id);
CREATE INDEX idx_orders_restaurant  ON orders(restaurant_id);
CREATE INDEX idx_order_items_order  ON order_items(order_id);

-- Index on date columns for time-series analysis
CREATE INDEX idx_order_date   ON orders(order_date);
CREATE INDEX idx_signup_date  ON customers(signup_date);
```

9.2 Query Optimization Techniques

- SELECT only required columns — Avoided SELECT * to reduce I/O and memory overhead.
- Filtered early with WHERE clauses — Applied conditions before aggregations to minimize working dataset.
- Used LIMIT strategically — Applied LIMIT to top-N queries to prevent full result set materialization.
- Replaced correlated subqueries with JOINs — JOIN operations outperform row-by-row subqueries on large tables.

9.3 CTE Usage

```
WITH revenue_by_restaurant AS (
  SELECT restaurant_id,
         SUM(order_amount - discount) AS net_revenue
  FROM orders
  GROUP BY restaurant_id
)
SELECT r.rest_name, rv.net_revenue
FROM revenue_by_restaurant rv
JOIN restaurants r ON rv.restaurant_id = r.restaurant_id
ORDER BY rv.net_revenue DESC;
```

9.4 Scalability Considerations

- Range partitioning the orders table by order_date improves query performance for time-series analysis at scale.
- Materialized views for pre-aggregated KPIs (monthly revenue, top restaurants) reduce dashboard query latency.
- A dedicated OLAP database (Redshift, BigQuery) would separate analytics workloads from production OLTP.

10. Key Findings Summary

The eight findings below represent the most significant and business-impactful conclusions from this SQL analysis.

1. Revenue Growth: 72.5% YoY Increase

Monthly revenue grew from ₹142K in January to ₹245K in December 2024 — a 72.5% annual growth rate, with October recording the highest monthly revenue at ₹231K (festive season effect). (Ref: Chart 1)

2. Metro Cities Drive 57.8% of Revenue

Mumbai, Bengaluru, and Delhi together generate 57.8% of total platform revenue, while Tier-2 cities show higher growth rates, signaling the next major growth frontier. (Ref: Charts 5, 9)

3. Customer Base Grew 309% in 3 Years

From 320 signups in Q1 2022 to 1,310 in Q4 2024, total platform customers grew by 309% — confirming strong and accelerating acquisition momentum. (Ref: Chart 2)

4. Top 3 Restaurants Control ~32% of Revenue

Spice Garden, Biryani House, and Pizza Palace collectively account for approximately 32% of total restaurant-generated revenue, indicating high value concentration. (Ref: Chart 4)

5. Delivery Time Gap: Mumbai vs Hyderabad (40%)

Mumbai's average delivery time of 42–50 minutes is up to 40% higher than Hyderabad's 29–37 minutes. This gap directly impacts customer satisfaction in the highest-revenue city. (Ref: Chart 3)

6. 37% Orders Use Discounts — Margin at Risk

Over one-third of all platform orders include a discount, representing a material reduction in gross revenue. Without clear LTV benefit from discounted users, this strategy requires immediate review. (Ref: Chart 11)

7. UPI Dominates at 42% — COD Declining

UPI is the preferred payment method at 42%. COD at 16% carries the highest operational cost and cancellation risk, making digital payment incentives a high-ROI investment. (Ref: Chart 8)

8. Biryani is the #1 Hero Product (4,820 Units)

Biryani leads all food items with 4,820 total units ordered — 17% more than the second-ranked item. It should be the centerpiece of all bundle deal and featured placement strategies. (Ref: Chart 6)

11. Business Recommendations

All recommendations below are directly derived from analytical findings and chart insights. Each is actionable, measurable, and linked to a specific business outcome

Revenue Growth

- Launch city-specific surge pricing during peak hours identified from monthly order trends (Chart 1) to maximize revenue per order without requiring volume increases.
- Introduce tiered commission structures for top-revenue restaurant partners, incentivizing them to list exclusive high-margin menu items on the platform.
- Expand aggressively into high-growth Tier-2 cities (Jaipur, Lucknow, Bhopal) identified in city revenue analysis (Chart 5) through targeted restaurant recruitment campaigns.

Customer Retention

- Design a VIP loyalty program for the top 10% of customers by spend — offering priority delivery, exclusive discounts, and early access to new restaurant listings.
- Implement gender-personalized marketing campaigns based on Chart 12 insights — weekday lunch promotions for female professionals and weekend family meal deals for male users.
- Launch a customer reactivation campaign for users inactive for 60+ days using personalized discounts based on their historical cuisine and restaurant preferences.

Cost Efficiency

- Restructure discount eligibility: target discounts at new customers only (first 3 orders), not habitual discount-seekers who inflate Chart 11's 37% usage rate without adding LTV.
- Transition COD-heavy markets to UPI/digital wallet incentives — 5% cashback on first UPI order — reducing the operational overhead of COD cancellations and cash handling.

Operational Improvement

- Deploy additional delivery agents in Mumbai (Chart 3's worst performer) prioritizing peak-hour coverage between 12:00–14:00 and 19:00–21:00 to reduce average delivery time below 40 minutes.
- Introduce a performance-based incentive structure for delivery agents: agents maintaining ratings above 4.5 and delivery times below city average receive monthly bonuses (supported by Chart 7 findings).
- Onboard at least 10 new restaurant partners per Tier-2 city to reduce average delivery distances and improve delivery time SLAs in emerging markets.

12. Conclusion

This project demonstrates the transformative power of structured SQL analysis when applied to real-world business data. By systematically querying a five-table relational database modeled after a leading online food delivery platform, we successfully converted raw transactional records into strategic business intelligence across six dimensions: revenue performance, geographic analysis, customer behavior, restaurant productivity, product analytics, and operational efficiency.

Twelve charts — spanning line charts, bar charts, and pie charts — provide visual validation of every key finding. From the 72.5% year-over-year revenue growth trend (Chart 1) and the 57.8% metro revenue concentration (Chart 9), to Biryani's dominance as the platform's hero product (Chart 6) and the 37% discount usage rate requiring immediate ROI review (Chart 11) — every visualization is directly supported by SQL query outputs and translates into concrete business action.

From a technical standpoint, this project showcases the full spectrum of SQL capabilities: aggregation, multi-table JOINs, subqueries, CTEs, date functions, and performance optimization through indexing and query design. These skills mirror exactly what is expected of a professional Data Analyst operating in the food-tech, e-commerce, or logistics domain.

13. Future Scope

The following enhancements represent natural evolution paths that would amplify the analytical impact of this project and enable real-time, predictive, and automated intelligence.

13.1 Real-Time Analytics

- Integrate Apache Kafka or AWS Kinesis to stream live order data, enabling real-time dashboards displaying current order volumes, revenue, and delivery performance.
- Build real-time alerting systems that trigger notifications when delivery time exceeds SLA thresholds or order cancellations spike in a city.

13.2 BI Tool Integration

- Connect the SQL database directly to Power BI, Tableau, or Metabase to create interactive visual dashboards extending and automating the charts built in this project.
- Build automated daily/weekly KPI email reports using scheduled SQL queries combined with Python (smtplib or SendGrid) delivering insights to leadership.

13.3 Machine Learning Integration

- Customer Churn Prediction: Train classification models using order frequency, recency, and AOV features to identify at-risk customers before they disengage.
- Demand Forecasting: Deploy time-series models (Prophet, ARIMA) trained on monthly order data (Chart 1) to forecast future demand and optimize staffing.
- Dynamic Pricing Engine: Develop a reinforcement learning model that adjusts delivery fees based on real-time demand, city-level delivery time (Chart 3), and supply signals.

13.4 Automation & Pipeline Development

- Build automated ETL pipelines using Apache Airflow or dbt to refresh analytical tables on a scheduled basis, eliminating manual query execution.
- Implement data quality monitoring using Great Expectations to validate incoming data against defined schemas before it enters the analytics layer.

13.5 Advanced Analytics

- Cohort Analysis: Track revenue and order frequency by customer acquisition cohort (signup month) to measure long-term marketing campaign effectiveness.
- Market Basket Analysis: Apply Apriori algorithm to order_items data to discover which food items are frequently ordered together for intelligent combo recommendations.
- Graph Analytics: Model the customer-restaurant-agent network to identify delivery routing optimization opportunities and network-driven growth effects.